Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Maurer, Richard H. Ph.D.		
Project Title:	Development of a Neutron Spectrometer to Assess Biological Radiation Damage		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	richard.maurer@jhuapl.edu	**Fax:**	FY 240-228-6696
PI Organization Type:	UNIVERSITY	**Phone:**	240-228-6482
Organization Name:	The Johns Hopkins University		
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City:	Laurel	**State:**	MD
Zip Code:	20723-6099	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	05/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Barghouty, N	**Contact Phone:**	256 544-0238
Contact Email:	Nasser.Barghouty@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1695		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The limited and specific aims of this task are 1) continued development of a portable neutron spectrometer based on existing prototypes; 2) characterize and calibrate a prototype flight instrument using high energy proton and heavy ion beams; 3) qualify the instrument for flight; 4) supplement the measurement of secondary charged particle spectra with the secondary neutron spectra produced at ground based accelerators related to the Lawrence Berkeley Laboratory materials research program. The objective is to complete the development of a flight qualified neutron energy spectrometer to be used to monitor the secondary neutron environment inside inhabitable spacecraft structures and on planetary surfaces. In ground based laboratory experiments the instrument will measure the secondary neutron environment produced when candidate spacecraft materials are irradiated with high energy heavy ion beams and protons. The high energy cosmic rays in interplanetary space collide with large spacecraft structures producing a greater number of energetic secondary neutrons inside the vehicle/habitat. Since there is no thick atmosphere, such as the one which		

	protects humans on the surface of the Earth, to absorb the high energy neutrons, these particles will penetrate deeply into astronauts' bodies, be slowed down efficiently by the water in the body and deposit large radiation doses in critical and blood forming organs. This process will increase the risk of carcinogenesis in vital organs for the astronauts significantly above that extant on the Earth's surface.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 04/27/2011)